MTH1002A

UNIVERSITY OF EXETER

FACULTY OF ENVIRONMENT, SCIENCE AND ECONOMY

MATHEMATICS AND STATISTICS

January 2024

Mathematical Methods

Module Leader: Dr L. Hakim

Duration: 2 HOURS

Answer Section A (50%) and any TWO of the three questions in Section B (25% for each).

Marks shown in questions are merely a guideline. Candidates are permitted to use approved portable electronic calculators in this examination.

This is a **CLOSED NOTE** examination.

No Word Count specified

SECTION A

1. (a) Find the inverse of the function

$$f(x) = \frac{x}{x^2 - 1} \quad \text{for} \quad |x| < 1.$$

(8)

(b) Use the ratio test to find the radius of convergence of the power series

$$\sum_{m=0}^{\infty} m^2 (27)^m x^{3m}.$$

(10)

(c) Suppose $z = (3 + i\sqrt{3})/5$. Find $|z|, \bar{z}, 1/z, \arg z$ and $\ln z$. Write z in polar form and use the result to determine z^{25}. (12)

(d) Compute $\|(2\vec{a}+\vec{b}) \times (\vec{a}+2\vec{b})\|^2$ for $\vec{a}, \vec{b} \in \mathbb{R}^3$ with $\|\vec{a}\| = 1, \|\vec{b}\| = 2$, and the angle between $\vec{a}$ and $\vec{b}$ equal to $2\pi/3$. (10)

(e) Find the derivative $\frac{dy}{dx}$ of the polar function $r = 1 + \sin\theta$ at $\theta = \frac{\pi}{4}$. (10)

[**50**]

SECTION B

2. (a) The functions $f(x), f_e(x)$ and $f_o(x)$ are defined by

$$f(x) = \frac{1}{1-x} = f_e(x) + f_o(x),$$

where f_e is an even function and f_o is an odd function.

(i) Find an expression for f_e and f_o. (4)

(ii) Sketch the three functions f, f_e and f_o. (6)

(b) (i) Derive the first three nonzero terms in the Maclaurin expansion of $\ln(1+x)$. (4)

(ii) Hence or otherwise, compute the limit

$$\lim_{x\to 0}\left[\frac{1-\cos x-\sin x+\ln(1+x)}{(1+x^3)^{1/2}-1}\right]$$

(6)

(c) Prove that the vector product of (x_1, x_2, x_3) and (y_1, y_2, y_3) is perpendicular to both these vectors. (5)

[25]

3. (a) Differentiate from first principles $f(x) = x^2 - 4x$. (6)

(b) For which values of $a, b \in \mathbb{R}$ does $\sum_{n=1}^{\infty} \frac{n^a}{b^n}$ converge or diverge? (8)

(c) In a triangle ABC, let the vectors $\overrightarrow{AB}, \overrightarrow{BC}, \overrightarrow{CA}$ be denoted by $\vec{u}, \vec{v}, \vec{w}$ respectively. Show that

$$\vec{u} \times \vec{v} = \vec{v} \times \vec{w} = \vec{w} \times \vec{u}.$$

(5)

(d) Consider the complex equation

$$\tan z = ia,$$

where a is a real constant.

(i) For what values of a does the equation have no solutions? (4)

(ii) For what values of a does the equation have an infinite number of solutions? (2)

[25]

4. (a) Let $a \neq 0$ be a real number.

(i) Describe the curve of the conic section given by

$$x^2 + y^2 - 2ax = 0. \qquad \text{(Eq1)}$$

(3)

(ii) Find the condition that m and c satisfy if the line $y = mx + c$ is tangent to the conic section described by (Eq1). (4)

(b) Show that

$$\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right).$$

(5)

(c) Suppose f is defined as below.

$$f(x) = \begin{cases} 2x - 3, & x < 5 \\ \frac{2}{3}x + b, & x \geq 5 \end{cases}$$

Determine the value of b so that $\lim_{x \to 5} f(x)$ exists. (5)

(d) Let A, B, C be the points $(1, 2, 0), (0, 1, 2), (2, 0, 1)$ in $\mathbb{R}^3$.

(i) Find a vector which is perpendicular to the plane containing A, B and C. (5)

(ii) Calculate the perpendicular distance from the point $(2, 2, 2)$ to this plane. (3)

[**25**]